Temporal Churn Analysis in SaaS: A Rolling Window Framework for Profit-Driven Retention

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Abstract

Customer churn prediction is critical for SaaS businesses aiming to retain users and maximize lifetime value. This paper introduces a rolling window framework that captures temporal patterns in customer behavior, enabling week-by-week churn predictions under realistic, evolving conditions. Beyond traditional churn defined by subscription cancellations, we investigate "quiet churn"—a reduction in usage that precedes cancellation, particularly relevant in SaaS where disengagement can go unnoticed. We incorporate Customer Lifetime Value (CLV) to estimate expected revenue loss, allowing prioritization of high-risk, high-value users. Our approach is evaluated on two datasets—one from a music platform and another from a SaaS firm—using metrics like F1-score, recall, and specificity. Results highlight the value of temporal modeling and financial weighting for proactive churn mitigation.

Keywords: *churn prediction, temporal modeling, rolling window, customer lifetime value, expected loss, SaaS, usage churn, quiet churn*

1 Introduction

In the competitive landscape of Software as a Service (SaaS) companies, customer retention is critical to driving sustainable growth and maintaining long-term profitability. As the cost of customer acquisition continues to rise, SaaS businesses must focus on improving retention, particularly by identifying users who are at risk of churning before they fully disengage. Accurate churn prediction models play a vital role in this effort, enabling organizations to proactively address potential loss and refine their customer success strategies.

However, traditional churn models, which often focus on explicit cancellation events—when a user formally terminates their subscription—miss a key aspect of churn behavior. However, this binary framing overlooks subtler patterns of disengagement that unfold over time. Jerath et al. (2011) highlighted that customers may "die" long before their final transaction or cancellation, emphasizing that churn can be unobserved yet behaviorally inferable. Usage churn often precedes cancellation, and in the context of SaaS, it can signal a much earlier opportunity for intervention. Identifying these early signs of disengagement is especially important for SaaS businesses, where customer retention is directly tied to recurring revenue and long-term profitability.

This study addresses a critical gap in the current literature by integrating temporal churn modeling with a focus on customer value alignment. Existing churn prediction models often fail to capture the nuanced and evolving nature of usage-based churn, and few incorporate business-relevant metrics such as Customer Lifetime Value (CLV) into their models. This paper proposes a novel framework that accounts for both temporal dynamics in user behavior and the financial value at risk, which enables a more holistic approach to churn prediction that is directly aligned with the business goals of a SaaS company.

By applying a rolling window framework, this study analyzes sequential, week-by-week activity data to detect shifts in user behavior over time. This provides a clearer picture of when and how churn signals emerge. Additionally, by integrating CLV, this framework not only identifies customers at risk but also ranks them based on the financial impact of their potential departure, allowing SaaS companies to prioritize retention efforts based on the highest-value customers. The model is tested on two real-world datasets: one from a music streaming service and another from a SaaS platform, offering insights into churn dynamics across different engagement models.

The findings from this study have significant implications for both academic research and business practice. By demonstrating the effectiveness of combining temporal churn analysis with value-at-risk metrics, this paper contributes to the growing body of knowledge on churn prediction in SaaS and offers practical guidance for improving retention strategies, ultimately helping SaaS companies maximize customer lifetime value and minimize churn-related revenue losses.

1.1 Literature Review

Customer churn prediction has emerged as a vital area of research due to its strategic importance in improving customer retention and profitability. In recent literature, a wide range of machine learning models have been developed and benchmarked, such as Bagging, Boosting (He and Ding, 2024) and hybrid architectures like Ensemble-Fusion (Lemmens and Croux, 2006). These models often report high levels of predictive accuracy. For example, Ensemble-Fusion achieved an F1 score of 96% (Lemmens and Croux, 2006), while BiLSTM-CNN models (Khattak et al., 2023) have demonstrated improved precision and recall through the integration of deep learning layers. Such advancements illustrate the technical progress in churn prediction techniques and signal a shift toward more complex and powerful classifiers.

However, a critical limitation of much of this literature is the lack of application to real-world industry data sets (He and Ding, 2024; Khattak et al., 2023). Many studies rely on synthetic or anonymized data; this limits the external validity and practical utility of their findings (He and Ding, 2024). This gap between academic experimentation and industry deployment creates challenges for assessing how well these models generalize across customer bases with varying behavior patterns, noise levels, and engagement contexts.

Furthermore, while high-accuracy metrics are frequently highlighted, many of these studies overlook the underlying feature diversity that fuels such predictions. In several cases, models are trained using narrow sets of numerical features or basic customer attributes, failing to incorporate richer behavioral signals, contextual data, or temporal dynamics. This restricts the model's ability to fully capture the complexity of churn behavior, especially in subscription-based platforms where disengagement can manifest subtly and gradually.

Moreover, few papers align model outputs with actionable business insights. Profitability-focused studies, such as those that aim to optimize campaign selection based on predicted churn risk, represent an important step in this direction Lemmens and Gupta (2020). However, even these often emphasize short-term gains without accounting for the long-term effects of retention interventions or the shifting nature of customer behavior over time.

In summary, while the literature presents numerous high-performing models, it suffers from three core limitations: limited testing on real-world datasets, insufficient feature diversity, and a lack of connection to operational retention strategies. Future work should emphasize robust feature engineering using diverse and longitudinal data, validate models on real customer activity datasets, and integrate predictive output with business impact metrics such as Customer Lifetime Value (CLV) or Expected Loss. Addressing these issues will enhance both the theoretical and practical relevance of churn prediction research.

2 Methodology

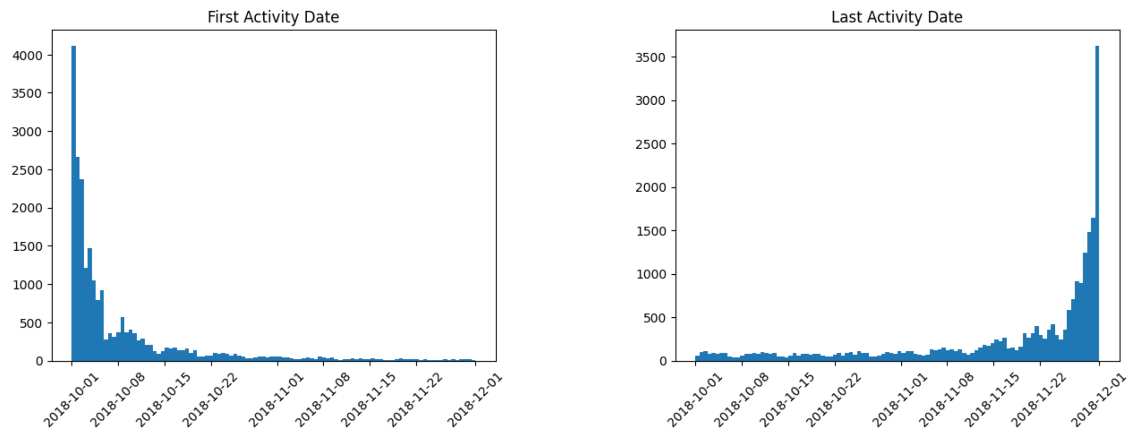
This section outlines the methodological framework used to develop and evaluate churn prediction models in a SaaS context. We begin by defining churn behavior. To capture customer behavior over time, we engineer a set of engagement-related features, including several temporal metrics. A rolling window algorithm is employed to simulate real-world prediction settings, where models are trained on sequential slices of longitudinal customer data. This enables us to account for changes in user behavior across time and better identify early signals of churn. We then train and compare multiple machine learning models, evaluating their performance using standard classification metrics. Finally, we introduce a profitability-based ranking system by applying an expected loss function, which estimates the financial impact of each potential churn event. This allows us to prioritize high-risk customers by value at risk, offering a more actionable and business-aligned output.

2.1 Data

This study utilizes two primary data sources to conduct a comprehensive analysis of churn prediction. The first is a a customer log dataset, sourced from Joy et al. (2024). This dataset comprises a 12.5 GB JSON file containing 18 columns and 26,259,199 records. The data encompasses a variety of user-related information, including user identifiers, interaction details, and technical details. The second data source is a proprietary dataset provided by an undisclosed SaaS company. This dataset contains information on customer interactions with the company's platform, including subscription details, product usage data, and customer service interactions.

2.2 Defining Churn

Churn can be defined in numerous ways. The datasets have click information that captures the when a subscription has been canceled. Based on the literature review conducted, many previous research have been conducted to predict churn as when a customer explicitly no longer wants to continue the service. However, many SaaS companies may benefit even with users who are not on a premium plan and use the service regularly. For instance, users may still allow SaaS companies to be profitable due to ad spend. Hence, it is valuable to measure customers who have "quietly churned" or stopped use the service.



(a) Displays when a customer begins using the platform.

(b) Displays when a customer stops using the platform.

Figure 1: Customer's propensity to churn between October and December.

To model behavioral churn in a temporal setting, we define churn as the absence of user activity in the current week following a week in which the user was active. This allows for identifying quiet churn, where a customer becomes inactive without formally canceling their subscription. Specifically, churn is operationalized at the user-week level using the following rule:

$$\text{Churn}_{i,t} = \begin{cases} 1, & \text{if Transactions}_{i,t} = 0 \text{ and Transactions}_{i,t-1} > 0 \\ 0, & \text{otherwise} \end{cases}$$

Where $\text{Churn}_{i,t}$ represents the churn status for user ii in week t, and $\text{Transactions}_{i,t}$ denotes the number of transactions made by the user in week t. A user is considered to have churned in a given week if they were active in the previous week (i.e., made at least one transaction) but had no activity in the current week. This definition supports a longitudinal analysis of churn by allowing customers to churn multiple times throughout their lifecycle, reflecting real-world usage behavior in subscription platforms.

2.3 Feature Engineering

To model user behavior over time, we process the dataset with rolling window calculations and weekly aggregation. The analysis focuses on understanding users' activity levels, specifically session counts and durations across various time windows. These features are valuable for predicting churn, as changes in engagement over time may indicate a shift toward quiet churn.

2.3.1 Rolling Window Calculations

We apply rolling window calculations to capture changes in user engagement across different time frames. These rolling windows (1, 2, 3, and 4 weeks) allow us to observe short-term fluctuations in user activity, which are essential for detecting changes in behavior that may indicate churn.

For each user, we compute the rolling sum of session counts over different window sizes $w \in \{7, 14, 21, 28\}$ (representing 1, 2, 3, and 4 weeks) and calculate the difference from the previous week:

$$S_w^{\text{rolling}}(t) = \sum_{i=t-w+1}^{t} S(i)$$

where $S(i)$ is the session count at time i, and w is the window size (in days).

Next, we compute the difference between the rolling sums for consecutive windows:

$$\Delta S_w(t) = S_w^{\text{rolling}}(t) - S_w^{\text{rolling}}(t-1)$$

This calculation allows us to track the change in user activity over time, which is useful for identifying trends in session counts.

2.3.2 Shifting for Previous Week Rolling Differences

Finally, we create lagged features for each rolling window by shifting the rolling differences to the previous week. This helps track how user activity changes from one week to the next:

$$\Delta S_w^{\text{prev}}(t) = \Delta S_w(t-1)$$

where $\Delta S_w^{\text{prev}}(t)$ represents the rolling difference from the previous week for window w, and we shift the value to align it with the current week's data.

This step ensures that we capture the temporal relationship between current and past user behavior, which is key for churn prediction.

2.3.3 Final Dataset Overview

After applying all the transformations, the final dataset `df_all_weeks` contains the following key features:

- **session_count**: The average number of items in session per user for each day.
- **session_duration**: The total session duration per user for each day.
- **dummy_level**: A categorical feature representing the user's level (e.g., premium, free).
- **prev_week_rolling_diff_X_days**: The rolling difference in session counts from the previous week for each window size.

These features provide valuable insights into user engagement and can be used to model churn by identifying significant changes in activity patterns over time.

3 Experimentation

To simulate real-time churn prediction in a production-like environment, we employed a rolling window framework for model training and evaluation. This strategy ensures that models only leverage historical data to predict future outcomes, mimicking how a deployed system would operate in practice.

For each week in the dataset (except the last), models were trained using data from that week and evaluated on the subsequent week. This sequential approach captures temporal drift and allows for robust assessment of model performance over time.

Given the imbalanced nature of churn classification (where churned users represent a small fraction of total users), we applied SMOTE (Synthetic Minority Over-sampling Technique; Chawla et al. (2002)) during training to balance the class distribution.

To evaluate performance, we used the F1-score, precision, recall, and specificity for each model on a weekly basis. Additionally, we ranked users by their predicted churn probability and calculated their *expected loss* using their Customer Lifetime Value (CLV). This allowed us to generate weekly exportable files of at-risk customers for potential retention targeting.

We experimented with the following machine learning models:

- Random Forest Classifier
- Logistic Regression
- Gradient Boosting Classifier (Friedman, 2001)
- XGBoost Classifier (Chen and Guestrin, 2016)
- LightGBM Classifier (Ke et al., 2017)
- Support Vector Machine (RBF Kernel)

3.1 Rolling Churn Prediction Algorithm

Let $W = \{w_1, w_2, \ldots, w_T\}$ represent the ordered set of weeks. For each week w_i, we define:

- X_{w_i}, y_{w_i}: Feature matrix and churn labels for week w_i.

Algorithm: Rolling Window Model Training and Testing

1. For each week index $i = 1$ to $T - 2$:
   (a) Set training week: $w_{\text{train}} = W_i$, and test week: $w_{\text{test}} = W_{i+1}$.
   (b) Extract data:
   $$X_{\text{train}} = X_{w_{\text{train}}}, \quad y_{\text{train}} = y_{w_{\text{train}}}, \quad X_{\text{test}} = X_{w_{\text{test}}}, \quad y_{\text{test}} = y_{w_{\text{test}}}$$
   (c) Apply SMOTE to balance training data:
   $$X'_{\text{train}}, y'_{\text{train}} = \text{SMOTE}(X_{\text{train}}, y_{\text{train}})$$
   (d) For each machine learning model M_k:
      i. Train the model:
      $$M_k \leftarrow M_k.\text{fit}(X'_{\text{train}}, y'_{\text{train}})$$
      ii. Predict labels for the next week:
      $$\hat{y}_{\text{test}} = M_k.\text{predict}(X_{\text{test}})$$
      iii. Evaluate using metrics such as F1-score, precision, recall, and specificity.
2. Aggregate weekly results to analyze model performance over time.

3.2 Customer Ranking by Expected Loss

Once a model has predicted the probability of churn for each customer, we combine this probability with the customer's estimated lifetime value (CLV) to calculate an *expected loss* value. This metric represents the potential financial impact if a particular customer churns. We define expected loss for user j as:

$$\text{ExpectedLoss}_j = P(\text{churn}_j) \cdot \text{CLV}_j$$

Where:

- $P(\text{churn}_j)$ is the model-predicted probability that user j will churn.
- CLV_j is the projected Customer Lifetime Value for user j.

In this study, Customer Lifetime Value is approximated based on historical revenue contributions from each user. Specifically, we estimate CLV_j as:

$$\text{CLV}_j = \text{MonthlyRevenue}_j \times \text{Tenure}_j$$

Where:

- MonthlyRevenue_j is the average monthly subscription fee or other revenue generated by user j.
- Tenure_j is the number of months user j has been active on the platform.

This approach captures the financial importance of each customer based on both how long they have remained subscribed and how much revenue they contribute on a recurring basis. By sorting customers in descending order of expected loss, we prioritize those who are not only likely to churn, but whose departure would also result in the greatest financial loss. This enables strategic, value-driven retention efforts.

4 Results

4.1 Model Performance Evaluation

The stacking classifier serves as a robust benchmark model to approximate the average performance of individual models over time. It offers a consolidated view of model effectiveness and helps determine when predictions become reliable. To facilitate a consistent evaluation of churn prediction models, we selected **Week 2024-11-11** as a representative benchmark.

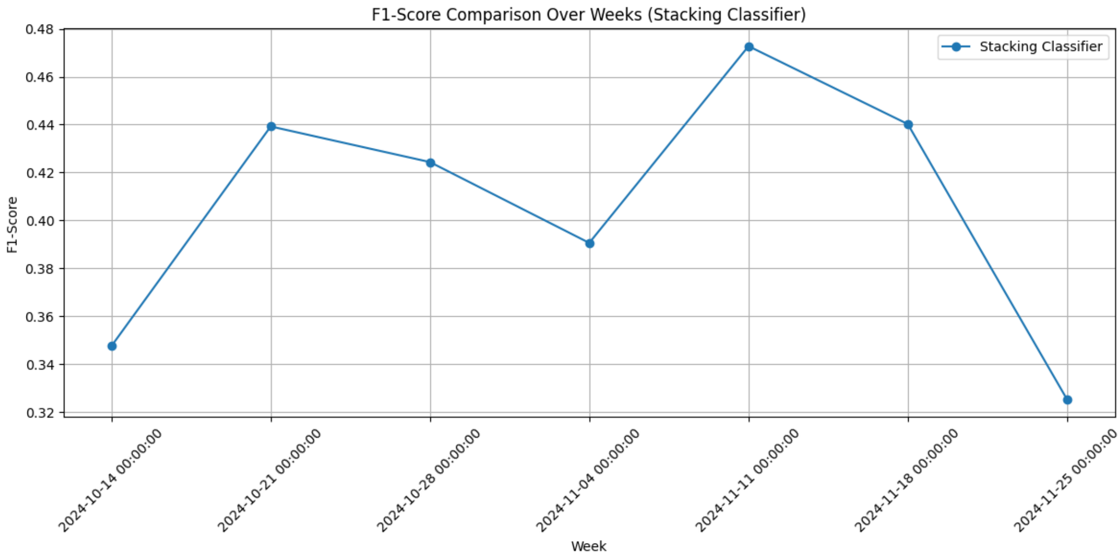
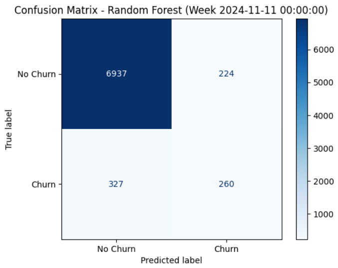


Figure 2: Performance of the stacking classifier over time, reflecting the aggregate effectiveness of individual models.

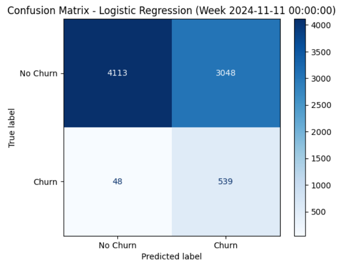
This week was chosen because it follows an initial four-week warm-up period, during which model performance was more volatile due to limited training data. As illustrated in Figure 2, by November 11th, model performance had begun to stabilize, reflecting improved generalization and more reliable predictions. This suggests that patterns in user behavior became more recognizable, allowing models to better distinguish between churn and retention cases. The models also appear to decline in performance after a certain period, potentially due to a diminishing relevance of historical data for predicting churn in real time beyond a specific timeframe. The following table summarizes precision, recall, and F1-score metrics for the `churn_label = 1` class (i.e., customers who churned), across multiple models on this selected week:

Model	Precision	Recall	F1-Score	Support
Random Forest	0.54	0.44	0.49	587
Logistic Regression	0.15	0.92	0.26	587
Gradient Boosting	0.20	0.63	0.31	587
Random Forest (Alt)	0.52	0.44	0.48	587
XGBoost	0.40	0.68	0.50	587
LightGBM	0.37	0.80	0.50	587
SVM	0.13	0.96	0.24	587

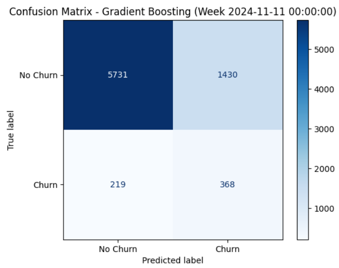
Table 1: Churn Prediction Metrics for Week 2024-11-04



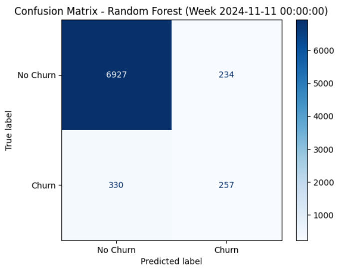
(a) Random Forest



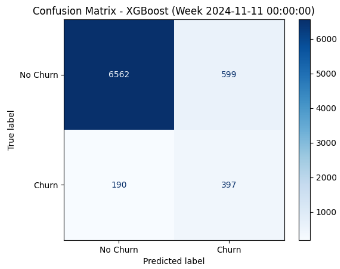
(b) Logistic Regression



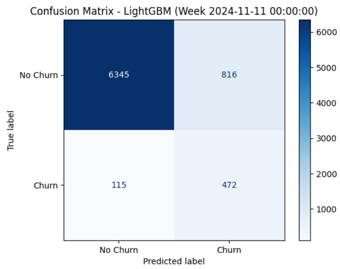
(c) Gradient Boosting



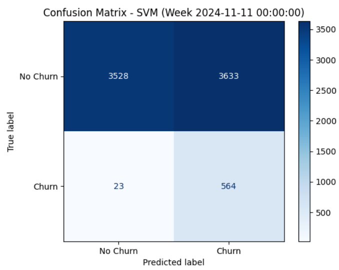
(d) Random Forest (Alt)



(e) XG Boost



(f) Light GBM



(g) SVM

Figure 3: F1-score trends over time for each model.

From the results, Gradient Boosting and XGBoost performed the most consistently, offering a balanced trade-off between precision and recall. Models such as Logistic Regression and SVM had high recall but suffered from very low precision, indicating a tendency to over-predict churn. In contrast, models like Random Forest and LightGBM provided more balanced outcomes.

These metrics highlight the importance of choosing models not solely based on recall but considering the F1-score to achieve a practical balance in churn detection tasks.

4.2 Expected Loss-Based Customer Ranking

To support targeted intervention strategies, we implemented an expected loss framework that combines each customer's predicted churn probability with their customer lifetime value (CLV).

This method enables prioritization of customers not just based on their risk of churn, but also the potential economic impact of their departure. The expected loss is calculated using:

$$\text{Expected Loss}_i = P(\text{churn}_i) \times \text{CLV}_i \tag{1}$$

where $P(\text{churn}_i)$ is the churn probability for customer i, and CLV_i is the corresponding customer lifetime value.

Customers are then ranked by their expected loss in descending order. This allows businesses to focus retention efforts on customers who present the highest potential financial loss if they were to churn.

Table 2 displays a sample output of this ranking for a specific week.

User ID	Week	Churn Risk	CLV	Expected Loss
104382	2024-11-11	0.92	$230.50	$212.06
108547	2024-11-11	0.85	$190.75	$162.14
107349	2024-11-11	0.78	$210.30	$164.03
105612	2024-11-11	0.65	$305.00	$198.25
103918	2024-11-11	0.70	$145.20	$101.64
109201	2024-11-11	0.60	$175.00	$105.00

Table 2: Sample Output of Ranked Customers by Expected Loss

Even with moderate model performance, ranking customers by churn risk and expected loss remains highly impactful. Rather than focusing solely on classification accuracy, businesses can use these rankings to prioritize retention efforts on high-risk, high-value customers. This approach ensures efficient allocation of resources and can yield significant returns, even when predictions are not perfect. The relative ordering of churn risk provides actionable insights that drive real business value.

5 Conclusion

This study explored week-by-week churn prediction in a SaaS context using a rolling window framework across multiple machine learning models. Among the models tested, LightGBM and XGBoost consistently achieved the highest F1-scores and recall, proving to be the most effective at identifying likely churners over time.

One of the most valuable contributions of this work is the use of the expected loss function, which ranks customers by combining churn risk with customer lifetime value (CLV). This ranking system provides a practical and business-oriented perspective. This allows for targeted retention efforts even when overall model accuracy is moderate.

The rolling window approach also provided insights into how model performance evolves over time. In the early weeks, models performed poorly due to limited training data. Performance stabilized after a 3–4 week "warm-up" period, but eventually declined again, suggesting diminishing returns from older historical data. This highlights a potential limitation in relying on static data or long-term histories for real-time churn prediction.

Despite encouraging results, several limitations remain. The F1-scores, while improving over time, indicate room for optimization, particularly in balancing precision and recall. Future work should include more extensive feature engineering to uncover latent signals in user behavior and platform interaction. Additionally, a new modeling strategy could be explored where only the most recent trailing weeks are used for training, rather than accumulating data from week 0 onward. This could help the model better capture short-term behavioral trends.

Another enhancement would be to train the model directly on expected loss as the target, rather than applying it as a post-processing step. This would align model optimization with business impact, potentially making predictions more actionable from the outset.

Finally, an important takeaway emerged when comparing these results to those reported in prior churn prediction literature. Many published models exhibit very high performance metrics, but they often rely on synthetic datasets or offline predictions. This research, by contrast, emphasizes the challenges and trade-offs of real-time, operational churn prediction using real user data, offering a more grounded and pragmatic view of model deployment in production environments.

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